SOCIAL MEDIA PLATFORM

PROJECT OVERVIEW

This project focuses on designing and analyzing a social media platform database using SQL.It includes creating relational tables such as users, posts, followers, likes, and comments.The system simulates real-world social media interactions and user activities. Advanced SQL concepts like joins, subqueries, views, triggers, and stored procedures are implemented.The project also performs data analysis to measure engagement, activity trends, and content performance.Overall, it demonstrates how SQL can be used for both database management and analytical insights.

OBJECTIVE:

To design and implement a basic social media database system that allows users to create posts, like posts, comment on posts, follow other users, and receive notifications. The system aims to manage user interactions efficiently and enable meaningful data analysis.

PROBLEM STATEMENT

Social media platforms generate large volumes of structured and unstructured data every day.Managing and analyzing this data efficiently is a major challenge for organizations.There is a need to store user interactions such as posts, likes, comments, and followers in a structured format.Without proper analysis, it is difficult to identify user engagement patterns and content performance.Businesses require insights such as top influencers, trending posts, and user activity trends for decision-making.Manual analysis of such data is time-consuming and prone to errors.There is also a need to automate processes like notifications and reporting within the system.This project addresses these challenges by using SQL to build, manage, and analyze a social media database effectively.

DATASET DESCRIPTION:

Users Table:

Column Name	Data Type	Description
user_id	INT	Primary Key
user_name	VARCHAR	User Name
email	VARCHAR	Email Address

Posts Table:

Column Name	Data Type	Description
post_id	INT	Primary Key
user_id	INT	Foreign Key
caption	TEXT	Post Caption
posted_at	DATETIME	Post Time

Comments Table:

Column Name	Data Type	Description
comment_id	INT	Primary Key
post_id	INT	Foreign Key
user_id	INT	Foreign Key
comment_text	TEXT	Comment Content
commented_at	DATETIME	Comment Time

Likes Table:

Column Name	Data Type	Description
like_id	INT	Primary Key
post_id	INT	Foreign Key
user_id	INT	Foreign Key
liked_at	DATETIME	Like Time

Followers Table:

Column Name	Data Type	Description
follower_id	INT	Foreign Key
following_id	INT	Foreign Key
follow_date	DATETIME	Follow Date

Notifications Table:

Column Name	Data Type	Description
user_id	INT	Foreign Key
message	TEXT	Notification Message
created_at	DATETIME	Created Time

Primary Key and Foreign Key Details:

Table Name	Primary Key	Foreign Key
Users	user_id	-
Posts	post_id	user_id
Comments	comment_id	post_id, user_id
Likes	like_id	post_id, user_id
Followers	-	follower_id, following_id
Notifications	-	user_id

Entity Relationship Diagram (ERD)

Users → Posts (1:M)
Users → Comments (1:M)
Posts → Comments (1:M)
Users → Likes (1:M)
Posts → Likes (1:M)
Users → Followers (M:N self relationship)
Users → Notifications (1:M)

ANALYSIS

Descriptive Analysis:

- The total number of users and posts were calculated to provide a clear overview of the platform's size, growth, and overall level of activity within the system.
- Engagement metrics such as the number of likes and comments per post were analyzed in detail to understand how users interact with content on the platform.
- Daily activity tracking was performed by grouping posts based on their posting dates, which helped in identifying patterns and trends in user posting behavior.
- Hashtag usage was examined carefully to determine the most frequently used hashtags and to understand the topics that are trending among users.
- Basic summaries of the dataset were generated using aggregate functions, which helped in gaining quick insights into user activity and content distribution.

Diagnostic Analysis:

- High engagement posts were identified and analyzed in depth to understand the key factors that contribute to increased likes and comments on certain posts.
- User activities were compared across different users to evaluate variations in participation, content creation, and overall engagement levels.
- Low engagement areas were investigated to determine possible reasons such as poor content quality, lack of visibility, or inactive user participation.
- Follower trends were analyzed to understand how the growth or decline in followers impacts user engagement and content reach.
- Data inconsistencies and anomalies were carefully examined and corrected to ensure that the analysis results are accurate and reliable.

Predictive Analysis:

- Historical engagement data was used to predict which types of posts are more likely to become trending in the future based on past performance patterns.
- User growth trends were analyzed to forecast the expected increase in the number of users joining the platform over time.
- Engagement patterns such as likes and comments were studied to estimate future interaction levels and user responsiveness.
- Active user behavior was analyzed to anticipate which users are likely to remain consistently active and contribute more content.
- Potential future influencers were identified by examining users with steadily increasing engagement and follower counts.

Prescriptive Analysis:

- Based on the insights derived from the data, effective content strategies were suggested to help users create more engaging and impactful posts.
- Optimal posting times were recommended by analyzing user activity patterns, which can help maximize reach and engagement.
- Improvements in engagement tactics were proposed, such as encouraging user interaction through comments, likes, and shares.
- Hashtag usage was optimized by recommending the use of trending and relevant hashtags to increase content visibility.
- Suggestions were provided to enhance overall user interaction on the platform, helping both users and administrators improve performance and engagement.

CONCLUSION

This project successfully demonstrates the creation and analysis of a social media platform using SQL. A well-structured relational database was designed with tables such as users, posts, followers, likes, and comments, ensuring proper relationships through primary and foreign keys. Various SQL concepts including joins, subqueries, views, triggers, and stored procedures were implemented to manage and analyze the data efficiently. The project also involved data cleaning, updates, and handling null values to maintain data consistency.

Through analytical queries, meaningful insights were generated such as identifying top influencers, analyzing post engagement, tracking user activity patterns, and detecting trending hashtags. These insights help in understanding user behavior and improving content strategies. Overall, the project highlights how SQL can be used not only for data storage but also for powerful data analysis, making it highly relevant for real-world applications in social media and business analytics.